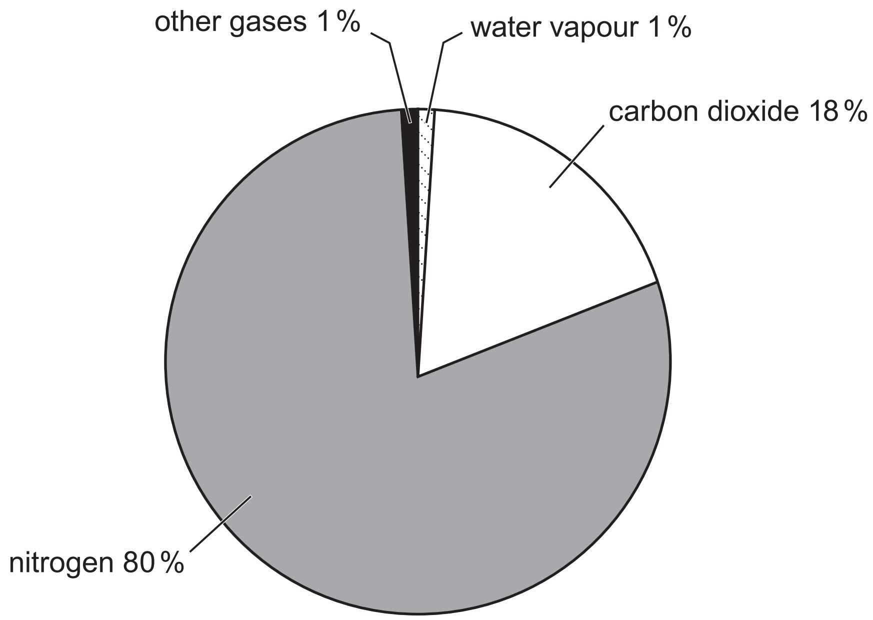
Examiner only

Answer **all** *questions.*

1. *(a)* The Earth's atmosphere has evolved over millions of years to its present composition.

The pie chart shows the composition of the atmosphere at one stage of its evolution.



(i) Describe the **main** differences in the composition of the atmosphere shown in the pie chart and that present on the Earth today. [3]

....

....

....

....

(ii) **Complete the table** to describe simple chemical tests that can be used in the laboratory to identify hydrogen and carbon dioxide. [2]

Gas	Test carried out	Expected observation
hydrogen		
carbon dioxide		

Examiner only

(b) During the last 250 years the level of carbon dioxide in the atmosphere has increased. Most scientists believe that the increase in the level of carbon dioxide in the atmosphere has resulted in global warming.

The table below shows some data about the atmosphere between 1750 and 2000.

Year	Concentration of carbon dioxide in the atmosphere (parts per million)	Average global temperature (°C)
1750	278	13.3
1800	282	13.4
1850	288	13.5
1900	297	13.7
1950	310	14.0
2000	368	14.6

Use the data in the table to answer parts (i) and (ii).

(i) Describe the trend in average global temperature between 1750 and 2000. [2]

....

....

....

(ii) Calculate the percentage increase in the concentration of carbon dioxide in the atmosphere from 1750 to 1850. [2]

percentage increase = ... %

9

3430UB01 03

Examiner only

Answer **all** *questions.*

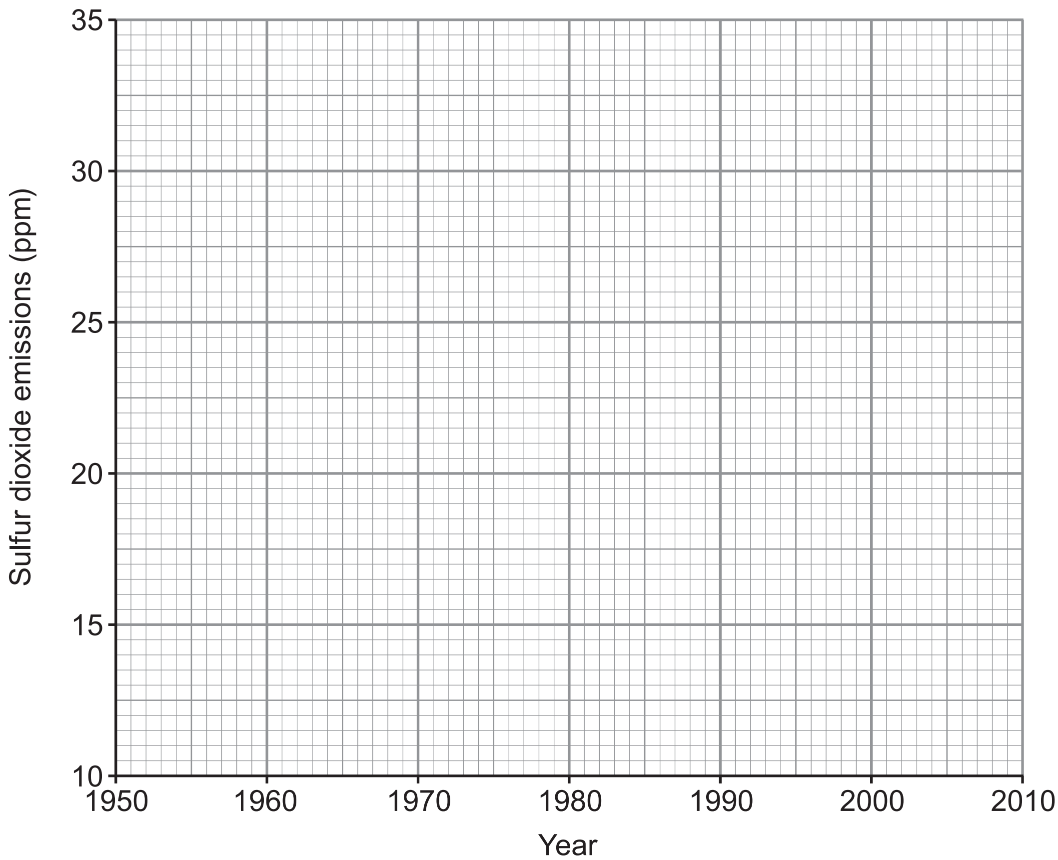
1. Burning fossil fuels containing sulfur causes sulfur dioxide, SO_2, to be released into the atmosphere.

The table shows sulfur dioxide emissions in the UK between 1950 and 2010.

Year	Sulfur dioxide emissions (ppm)
1950	12.0
1960	16.0
1970	21.5
1980	29.5
1990	29.0
2000	24.0
2010	18.5

ppm = parts per million

(a) (i) On the grid plot the sulfur dioxide emissions against the year and draw a suitable line. [3]



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(ii) Describe how sulfur dioxide emissions changed between 1950 and 2010. [2]

....

....

....

(iii) The UK government introduced a regulation to reduce sulfur dioxide emissions in the 1980s. From your graph, state why it is difficult to decide exactly the year when the regulation came into force. [1]

....

....

....

(b) Sulfur dioxide can be converted to sulfur and water by reacting it with hydrogen sulfide, H_2S.

Complete and balance the symbol equation for this reaction. [2]

$$SO_2 + \boxed{}\, H_2S \longrightarrow \boxed{}\, S + \boxed{}\, \dots\dots\dots$$

3430UB01 03

8

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(iii) Put a tick (✓) in the box next to the **two** statements which describe the conclusions that can be drawn from the data in **Figure 1**. [2]

lead contamination in road-side soil decreases in towns and in the countryside as the distance from the centre of the road increases ☐

the decrease in lead contamination between 12 m and 42 m from the centre of the road is greater in towns than in the countryside ☐

lead contamination in road-side soil at all distances is much greater in towns than in the countryside ☐

lead contamination in road-side soil decreases between 12 m and 42 m from the centre of the road in the countryside ☐

there is approximately 50 % more lead contamination in road-side soil in towns compared to the countryside between 12 m and 28 m from the centre of the road ☐

there is an overall decrease of nearly 70 % in the lead contamination of road-side soil in towns between 12 m and 42 m from the centre of the road ☐

(iv) In the mid-1970s the use of lead in paints and the manufacture of cars that used leaded petrol was banned. Put a tick (✓) in the box next to the statement which **best** describes what happened after the ban. [1]

paint and petrol were lead-free by the mid-1970s anyway ☐

paint and petrol were not lead-free until 1980 ☐

the use of lead-based paint and leaded petrol increased until 1980 before decreasing ☐

it took 15 years for paint and petrol to become lead-free ☐

Examiner only

(b) (i) Lead can form a number of oxides with different formulae. Red lead, Pb_3O_4, is used to make batteries. It is manufactured by reacting lead carbonate with oxygen. Carbon dioxide is also produced.

Balance the equation for this reaction. [1]

$$\boxed{}\,PbCO_3 + O_2 \longrightarrow \boxed{}\,Pb_3O_4 + \boxed{}\,CO_2$$

(ii) 11.36 g of a lead oxide contains 10.18 g of lead. Calculate the empirical formula of this oxide. [3]

$A_r(Pb) = 207$ $A_r(O) = 16$

Empirical formula ....

9

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
							1 H Hydrogen 1										4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3430UB0-1

S19-3430UB0-1

WEDNESDAY, 12 JUNE 2019 – MORNING

SCIENCE (Double Award)

Unit 2: CHEMISTRY 1
HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**8**	
2.	**7**	
3.	**8**	
4.	**9**	
5.	**5**	
6.	**6**	
7.	**9**	
8.	**8**	
Total	**60**	

3430UB01
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

JUN193430UB0101

Examiner only

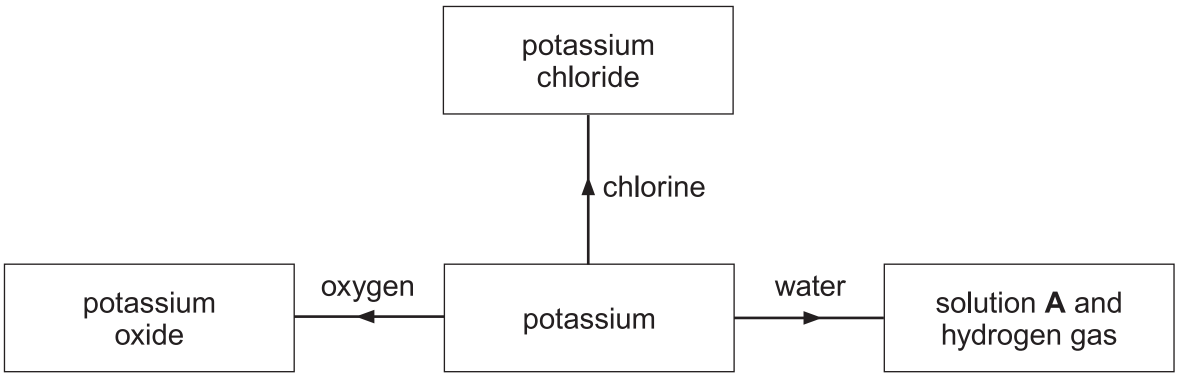
Answer **all** *questions.*

1. *(a)* The table gives information about some elements.

Element	Electronic structure	Group	Period
oxygen	2,6	6	2
chlorine	....	7	3
....	2,8,5	5	3
potassium	2,8,8,1	1	....

Complete the table. [3]

(b) The flow chart shows some of the reactions of potassium.



(i) State **one** observation you would make when potassium reacts with water. [1]

....

....

(ii) Apart from wearing gloves and safety goggles, give **one** safety precaution that should be taken when adding potassium to water. [1]

....

....

Examiner only

(iii) Give the formula of solution **A**. [1]

....

(iv) Suggest a value for the pH of solution **A**. [1]

....

(v) Name a Group 1 metal that is **more** reactive than potassium. [1]

....

8

3430UB01 03

Examiner only

Answer **all** questions.

1. (a) (i) State why sodium is stored in oil in the laboratory. [1]

....

(ii) Describe the change in appearance when a piece of freshly cut sodium is left for a few minutes. [1]

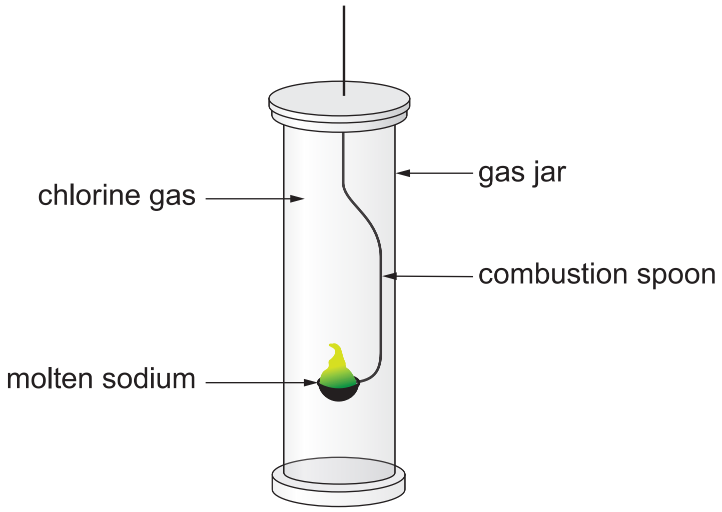
....

....

(iii) Give the formula of the compound formed when sodium reacts with oxygen. [1]

....

(b) The diagram shows the reaction of sodium with chlorine.



(i) State why it is necessary to carry out this reaction in a fume cupboard. [1]

....

(ii) Complete and balance the equation for the reaction of sodium and chlorine. [2]

$\square$ Na + Cl_2 $\longrightarrow$ $\square$ ....

Examiner only

(c) The table shows some properties of Group 7 elements.

Element	Melting point (°C)	Boiling point (°C)	Reaction with hot iron
fluorine	–220	–188	explosive
chlorine	–101	–34	very fast
bromine	–7	59	quite fast
iodine	114		slow

(i) Put a tick (✓) in the box next to the most likely boiling point for iodine. [1]

–25 °C ☐ 25 °C ☐ 100 °C ☐ 150 °C ☐

(ii) Astatine lies below iodine in Group 7. State how you would expect astatine to react with hot iron.

Give a reason for your answer. [2]

....

....

....

3430UB01
03

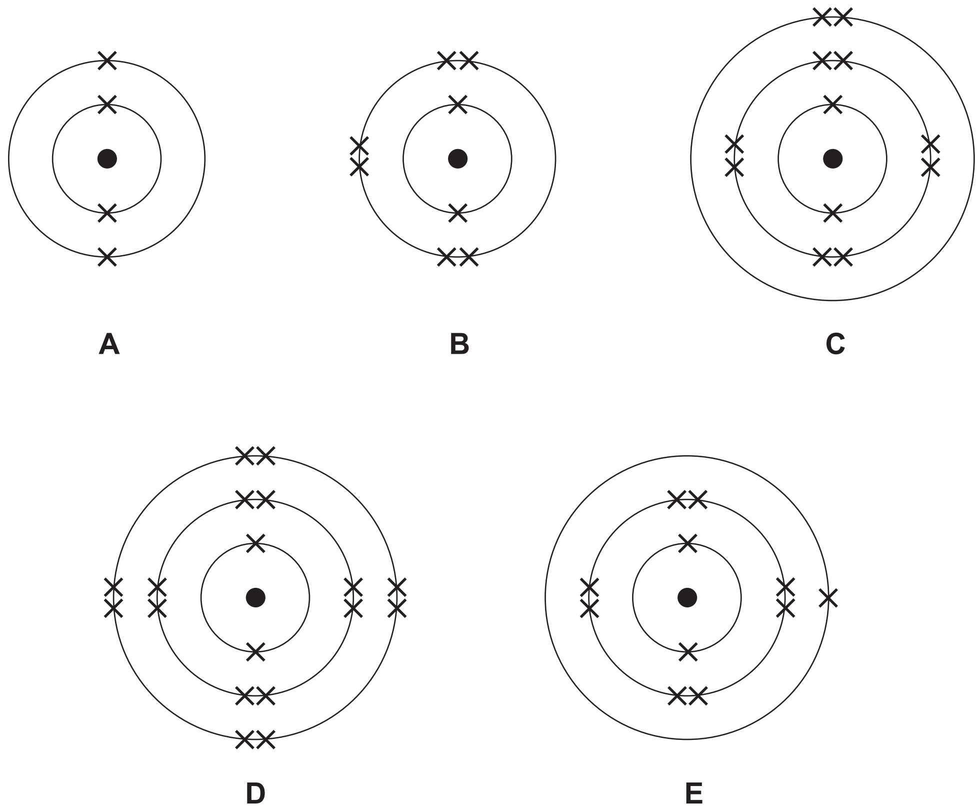
9

Examiner only

Answer **all** questions.

1. The diagrams below show the electronic structures of five elements, **A**, **B**, **C**, **D** and **E**.

The letters are not the symbols of the elements.



(a) Give the **letters** of the elements found in Period 2 of the Periodic Table. Give a reason for your choice. [2]

Letters .... and ....

Reason ....

....

 (3430UB0-1)

Examiner only

(b) Give the **letter** of the element found in Group 0 of the Periodic Table. Give a reason for your choice. [2]

Letter ....

Reason ....

....

(c) Explain how the electronic structure of element **E** can be used to determine its atomic number. [2]

....

....

....

(d) One of the diagrams represents the element oxygen. Oxygen reacts with potassium to form potassium oxide.

Give the formula for potassium oxide and balance the equation for this reaction. [2]

$$\boxed{}\,K + O_2 \longrightarrow 2\ \text{..}$$

3430UB01 03

8